



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

NATIONAL SENIOR CERTIFICATE EXAMINATIONS
SENIORSERTIFIKAAT-EKSAMEN/
NASIONALE SENIORSERTIFIKAAT-EKSAMEN

GRADE/GRAAD 12

TECHNICAL MATHEMATICS P1/TEGNIESE WISKUNDE V1

2022

FINAL MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

Marking Codes/Nasienkode	
A	Accuracy/Akkuraatheid
CA	Consistent Accuracy/Volgehoue Akkuraatheid
M	Method/Metode
R	Rounding/Afronding
NPR	No Penalty for Rounding/Geen Penaliserings vir Afronding nie
NPU	No Penalty for Units omitted/Geen Penaliseering vir Eenhede Weggelaat nie
S	Simplification/Vereenvoudiging
SF	Substitution in Correct Formula/Vervanging in Korrekte Formule
AO	Answer only/ Slegs antwoord

These marking guidelines consist of 21 pages.
Hierdie nasienriglyne bestaan uit 21 bladsye.

NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- Consistent accuracy applies to all aspects of the marking guidelines where indicated.
- If a candidate strikes off a response to a question and does not attempt the question again, then the struck off question should be marked.

LET WEL:

- Indien 'n kandidaat 'n vraag TWEE keer beantwoord, sien slegs die EERSTE poging na.
- Volgehoue akkuraatheid is deurgaans op alle aspekte van die nasienriglyne van toepassing.
- Indien 'n kandidaat 'n antwoord deurhaal en nie poog om die vraag weer te beantwoord dan moet die deurgehaalde antwoord gemerk word.

QUESTION/VRAAG 1

1.1.1	$x^2 - 3x - 10 = 0$ $(x - 5)(x + 2) = 0$ OR/OF $x = \frac{-(-3) \pm \sqrt{(-3)^2 - 4(1)(-10)}}{2(1)}$ $x = -2$ or/of $x = 5$	✓ factors/ faktore or/of SF A ✓ both values /beide waardes van x CA (2)
1.1.2	$x^2 - 3x - 10 < 0$ $(x - 5)(x + 2) < 0$ $-2 < x < 5$ OR/OF $x \in (-2; 5)$ OR/OF $x > -2$ and/en $x < 5$  Note: If correct number line representation is shown and where no other calculations are shown then Award full marks. <i>Nota: Indien slegs die korrekte getallelyn en geen ander berekeninge vertoon word kan volpunte toegeken word.</i>	✓ critical values/kritiese waardes CA ✓ correct notation/korrekte notasie A ✓ number line representation/ Getallelyn voorstelling CA (3)
1.2	$2x^2 - 11 = -7x$ $2x^2 + 7x - 11 = 0$ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $x = \frac{-7 \pm \sqrt{7^2 - 4(2)(-11)}}{2(2)}$ $x = -4,68$ or/of $x = 1,18$	✓ standard form/standaardvorm A ✓ SF CA ✓ both values of x correctly rounded/ beide waardes van x korrek afgerond CA

OR/OF

$$0 = -2x^2 - 7x + 11$$

$$\therefore x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(-2)(11)}}{2(-2)}$$

$$\therefore x = -4,68 \text{ or/of } x = 1,18$$

Penalty for Rounding/ Penalisering vir Afronding

OR/OF

✓ standard form/standaardvorm **A**

✓ SF **CA**

✓ both values of x correctly rounded/
beide waardes van x korrek afgerond **CA**
(3)

1.3

$$y - x + 1 = 0 \text{1}$$

$$y + 7 = x^2 + 2x \text{2}$$

$$y = x - 1$$

$$(x - 1) + 7 = x^2 + 2x$$

$$\therefore x^2 + x - 6 = 0$$

$$(x + 3)(x - 2) = 0 \text{ OR/OF } x = \frac{-(1) \pm \sqrt{(1)^2 - 4(1)(-6)}}{2(1)}$$

$$x = -3 \text{ or/of } x = 2$$

$$y = -3 - 1 \text{ or/of } y = 2 - 1$$

$$y = -4 \text{ or/of } y = 1$$

OR/OF

$$x = y + 1$$

$$y + 7 = (y + 1)^2 + 2(y + 1)$$

$$y + 7 = y^2 + 2y + 1 + 2y + 2$$

$$y^2 + 3y - 4 = 0$$

$$(y + 4)(y - 1) = 0 \text{ or/of } y = \frac{-(3) \pm \sqrt{(3)^2 - 4(1)(-4)}}{2(1)}$$

$$\therefore y = -4 \text{ or/of } y = 1$$

$$x = -4 + 1 \text{ or/of } x = 1 + 1$$

$$\therefore x = -3 \text{ or/of } x = 2$$

OR/OF

✓ x subject of formula/onderwerp van
formule **A**
✓ substitution/vervanging **CA**

✓ correct standard form/korrekte
standaardvorm **CA**

✓ factors/ faktore or/of SF **CA**

✓ both/ beide y-values/-waardes **CA**

✓ both/ beide x-values/-waardes **CA**

(6)

1.4.1	$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$ $\frac{1}{R_p} = \frac{R_1 + R_2}{R_1 \cdot R_2}$ $R_p = \frac{R_1 \cdot R_2}{R_2 + R_1}$ OR/OF $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$ $R_p \left(\frac{1}{R_p} \right) = R_p \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$ $R_p = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$	✓ simplification/vereenvoudiging A ✓ subject/onderwerp CA OR/OF ✓ simplification/vereenvoudiging A ✓ subject/onderwerp CA AO: full marks/volpunte (2)
1.4.2	$R_p = \frac{40 \times 45}{45 + 40} = \frac{1800}{85}$ $R_p = \frac{360}{17} \Omega \quad \textbf{OR/OF} \quad 21,18 \Omega$ OR/OF $\frac{1}{R_p} = \frac{1}{40} + \frac{1}{45} = \frac{17}{360}$ $R_p = \frac{360}{17} \Omega \quad \textbf{OR/OF} \quad 21,18 \Omega$ OR/OF $R_p = \frac{1}{\frac{1}{40} + \frac{1}{45}}$ $R_p = \frac{360}{17} \Omega \quad \textbf{OR/OF} \quad 21,18 \Omega$	✓ SF CA ✓ value of/waarde van R_p CA OR/OF ✓ SF A ✓ value of/waarde van R_p CA OR/OF ✓ SF CA ✓ value of/waarde van R_p CA AO: full marks/volpunte (2)

1.5	1101100₂ ÷ 1100₂ = 1001₂ <table><tr><td>2⁶</td><td>2⁵</td><td>2⁴</td><td>2³</td><td>2²</td><td>2¹</td><td>2⁰</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td></td><td></td><td></td><td>1</td><td>1</td><td>0</td><td>0</td></tr></table> 64 + 32 + 8 + 4 = 108 8 + 4 = 12 ∴ 108 ÷ 12 = 9 9 = 1001₂ OR/OF 1001 1100 1101100 1100 11 0 110 0 1100 1100 1101100₂ ÷ 1100₂ = 1001₂	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	1	1	0	1	1	0	0				1	1	0	0	✓ M A ✓ 1001₂ CA OR/OF ✓ M A ✓ 1001₂ CA AO: full marks/volpunte (2) [20]
2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰																	
1	1	0	1	1	0	0																	
			1	1	0	0																	

QUESTION/VRAAG 2

2.1.1	$e = 4$	✓ Value of/Waarde van e	A (1)
2.1.2	$e < 4$	✓ Value of/Waarde van e	A (1)
2.2	$mx^2 - 12x + 9 = 0$ $\Delta = b^2 - 4ac$ $\Delta = (-12)^2 - 4(m)(9)$ $\Delta = 144 - 36m$ For equal roots /Vir gelyke wortels: $\Delta = 0$ $144 - 36m = 0$ $m = 4$	✓ SF ✓ S ✓ $\Delta = 0$ ✓ Value of/Waarde van m	A CA A CA (4) [6]

QUESTION/VRAAG 3

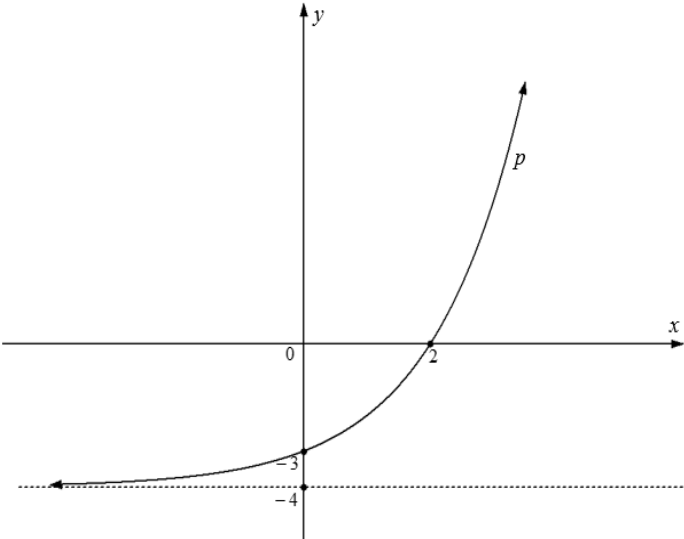
3.1.1	$\frac{3^x \times 3^{x-2}}{9^{x-3}}$ $= \frac{3^x \times 3^{x-2}}{3^{2x-6}}$ $= 3^{x+x-2-2x+6}$ $= 3^4 = 81$ OR/OF $\frac{3^x \times 3^{x-2}}{9^{x-3}}$ $= \frac{3^x \times 3^{x-2}}{3^{x-3} \times 3^{x-3}}$ $= 3^{x+x-2-x+3-x+3}$ $= 3^4 = 81$ OR/OF $\frac{3^x \times 3^{x-2}}{9^{x-3}}$ $= \frac{3^x \times 3^x \times 3^{-2}}{9^x \times 9^{-3}}$ $= \frac{9^x \times 9^{-1}}{9^x \times 9^{-3}}$ $= 9^2 = 81$	✓ prime base/<i>priembasis</i> 3^2 A ✓ exponent properties/<i>eksponent- eienskap</i> CA ✓ 3^4 or/of 81 CA OR/OF ✓ prime base/<i>priembasis</i> A ✓ exponent properties/<i>eksponent- eienskap</i> CA ✓ 3^4 or/of 81 CA OR/OF ✓ Split powers /<i>Verdeel magte</i> A ✓ exponent properties/<i>eksponent- eienskap</i> CA ✓ 9^2 or/of 81 CA (3)
3.1.2	$(\sqrt{5} + 4)^2 - \sqrt{45}$ $= 5 + 8\sqrt{5} + 16 - 3\sqrt{5}$ $= 21 + 5\sqrt{5}$	✓ squaring binomial/<i>kwadreer binoom</i> A ✓ $3\sqrt{5}$ A ✓ $21 + 5\sqrt{5}$ CA (3)

3.1.3	$\log_{32} 8 + \log 10$ $= \frac{\log 8}{\log 32} + 1$ $= \frac{\log 2^3}{\log 2^5} + 1$ $= \frac{3 \log 2}{5 \log 2} + 1$ $= \frac{3}{5} + 1$ $= \frac{8}{5} = 1,6$	✓ change of base law/verandering van basis wet A ✓ 1 A ✓ log property/-eienskap A ✓ $\frac{8}{5}$ or/of 1,6 CA AO: zero marks/punte (4)
3.2	$\log_4 x + \log_4 (x - 6) = \log_5 25$ $\log_4 x(x - 6) = \log_5 5^2$ $\log_4 (x^2 - 6x) = 2$ $x^2 - 6x = 4^2$ OR / OF $\log_4 (x^2 - 6x) = \log_4 4^2$ $x^2 - 6x - 16 = 0$ $(x - 8)(x + 2) = 0$ OR / OF $x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(-16)}}{2(1)}$ $x = 8$ or/of $x \neq -2$ $\therefore x = 8$	✓ $\log_4 x(x - 6)$ A ✓ 2 A ✓ log definition/prop./-definisie/-eiensk. CA ✓ factors/ faktore or/of SF CA ✓ $x = 8$ CA ✓ $x \neq -2$ CA (6)
3.3.1	$z_1 = 4\sqrt{2} \text{ cis } (225^\circ)$ $= 4\sqrt{2} \cos 225^\circ + 4\sqrt{2} \sin 225^\circ \cdot i$ $= -4 - 4i$	✓ polar expansion/polêre uitbrei A ✓ $-4 - 4i$ CA AO: full marks/volpunte (2)
3.3.2	$z_1 + z_2$ $= (-4 - 4i) + (3 - 4i)$ $= -4 - 4i + 3 - 4i$ $= -1 - 8i$	✓ Substitution/Vervanging CA ✓ $-1 - 8i$ CA (2)

3.4	$-p + qi = 4i^5 - 2(7 + 3i)$ $-p + qi = 4(i^2)^2 i - 14 - 6i$ $-p + qi = 4(-1)^2 i - 14 - 6i$ $-p + qi = 4i - 14 - 6i$ $-p + qi = -14 - 2i$ $\therefore p = 14 \text{ and/en } q = -2$	✓ rewriting/herskryf i^5 as $(i^2)^2 i$ A ✓ replacing/vervang i^2 with/met -1 A ✓ S CA ✓ p value/-waarde CA ✓ q value/-waarde CA (5) [25]
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QUESTION/VRAAG 4

4.1.1	$f(x) = \frac{a}{x} + 2$ $4 = \frac{a}{-2} + 2$ $\frac{a}{2} = -2$ $a = -4$ $\therefore f(x) = -\frac{4}{x} + 2$	✓ substituting/vervang. 2 A ✓ subst./verv. $(-2; 4)$ A ✓ value of/waarde van a CA
4.1.2	$g(x) = x + 2$	✓ equation/vergelyking A (1)
4.1.3	$y \in \mathbb{R}, y \neq 2$ OR/OF $y \in (-\infty; 2) \cup (2; \infty)$ OR/OF $y > 2$ or/of $y < 2$	✓ critical value/kritiese waarde A ✓ notation/notasie A OR/OF ✓ critical value/kritiese waarde A ✓ notation/notasie A OR/OF ✓ critical value/kritiese waarde A ✓ notation/notasie A (2)
4.1.4	$f(x) = -\frac{4}{x} + 2$ $0 = -\frac{4}{x} + 2$ $\frac{4}{x} = 2$ $2x = 4$ $x = 2$ OR/OF $W(2; 0)$	✓ $0 = -\frac{4}{x} + 2$ A ✓ x-coordinate of/koördinaat van W CA OR/OF ✓ x-coordinate /koördinaat A ✓ y-coordinate /koördinaat A (2)

4.1.5	$g(x) = 2 + 2 = 4$ $\therefore V(2; 4)$ OR/OF $\therefore V(2; 4)$	✓ substitution of x-value of W CA ✓ coordinates of/koördinate van V CA OR/OF ✓ x-coordinate /koördinaat A ✓ y-coordinate /koördinaat A (2)
4.2.1	$y = -4$	✓ equation/vergelyking A (1)
4.2.2	$p(x) = 2^x - 4$ y-intercept/-afsnit; $x = 0$ $p(0) = 2^0 - 4$ $\therefore y = -3$ x- intercept/-afsnit; $y = 0$ $0 = 2^x - 4$ $2^x = 2^2$ OR/OF $2^x = 4$ $\therefore x = 2$ $\therefore x = \log_2 4$ $\therefore x = 2$	✓ $y = -3$ A ✓ $2^x = 2^2$ or/of $x = \log_2 4$ A ✓ $x = 2$ CA AO: full marks/volbunte (3)
4.2.3		✓ shape/vorm A ✓ both intercepts/beide afsnitte CA ✓ asymptote/asimptoot CA (3)
4.3.1	$k = (4)^2 - 2(4) + 4$ $\therefore k = 12$	✓ substitution/vervanging A ✓ value of/waarde van k CA AO: full marks/volbunte (2)

4.3.2	$g(x) = x + 1$ $C(4; y)$ $y_C = 4 + 1 = 5$ OR / OF $AC = \sqrt{(x_2 - x_1)^2 - (y_2 - y_1)^2}$ $AC = y_A - y_C$ $= 12 - 5$ $= 7 \text{ units/eenhede}$ OR/OF $AC = (x^2 - 2x + 4) - (x + 1)$ $= x^2 - 3x + 3$ $= (4)^2 - 3(4) + 3$ $= 7 \text{ units/eenhede}$	 ✓ y_C value/-waarde A ✓ M A ✓ length of/lengte van AC CA OR/OF ✓ M A ✓ substitution/vervanging CA ✓ length of/lengte van AC CA (3)
4.3.3	$f(x) = x^2 - 2x + 4$ $x = -\frac{b}{2a}$ OR/OF $f'(x) = 0$ $x = -\frac{(-2)}{2(1)}$ $2x - 2 = 0$ $x = 1$ $x = 1$ $\therefore \frac{x_p + 4}{2} = 1$ $\therefore x_p = -2$ $f(-2) = (-2)^2 - 2(-2) + 4 = 12$ $\therefore P(-2; 12)$ OR/OF $f(x) = x^2 - 2x + 4$ $P(x; 12)$ $12 = x^2 - 2x + 4$ $x^2 - 2x - 8 = 0$ $(x - 4)(x + 2) = 0$ OR / OF $x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-8)}}{2(1)}$ $\therefore x = 4 \text{ or/of } x = -2$ $\therefore P(-2; 12)$	 ✓ SF or/of $2x - 2 = 0$ A ✓ value of /waarde van x CA ✓ value of/ waarde van x_p CA ✓ coordinate/koördinaat CA OR/OF ✓ equating/gelykstelling CA ✓ standard form/standaardvorm CA ✓ factors/ faktore or/of SF CA ✓ $x = -2$ CA (4) [26]

Note: in Q5 the use of wrong formula must be regarded as a breakdown

Nota: in V5 moet die gebruik van verkeerde formule as 'n afbreek

QUESTION/VRAAG 5

5.1	$A = P(1 + in)$ $A = R15350 \left(1 + \frac{21}{100} \times 2 \right)$ $A = R21\,797$ Monthly payment/ Maandelikse paaieiment $= \frac{R21\,797}{24} = R908,21$ OR/OF $SI = P \times i \times n$ $SI = 15350 \times 21\% \times 2 = R6\,447$ $\therefore A = 15350 + 6\,447 = R21\,797$ Monthly payment/ Maandelikse paaieiment $= \frac{R21\,797}{24} = R908,21$	✓ SF A ✓ value of/waarde van A CA ✓ M divide by/deel deur 24 CA ✓ monthly payment/maandelikse paaieiment CA OR/OF ✓ SF A ✓ value of/waarde van A CA ✓ M divide by/deel deur 24 CA ✓ monthly payment/maandelikse paaieiment CA NPR (4)
5.2.1	$500 \times 1,25 = 625$	✓ value/waarde A (1)
5.2.2	$A = P(1 + i)^n$ $= 500(1 + 0,25)^{10}$ $= 4656,61$ ≈ 4656	✓ F A ✓ SF A ✓ value of/waarde van CA (3)
5.3	$A = P(1 + i)^n$ $50962,58 = P \left(1 + \frac{6,78\%}{12} \right)^{24} \left(1 + \frac{5,20\%}{4} \right)^6$ $P = \frac{50962,58}{\left(1 + \frac{6,78\%}{12} \right)^{24} \left(1 + \frac{5,20\%}{4} \right)^6}$ $\approx R41197,43$	✓ 24 A ✓ $\left(1 + \frac{6,78\%}{12} \right)$ A ✓ 6 A ✓ $\left(1 + \frac{5,20\%}{4} \right)$ A ✓ P the subject/die onderwerp CA ✓ value/waarde CA

OR/OF

$$A = P(1+i)^n$$

$$A_2 = P \left(1 + \frac{6,78\%}{12} \right)^{24}$$

$$= (1,144\dots)P$$

$$A_{3,5} = 1,145P \left(1 + \frac{5,20\%}{4} \right)^6$$

$$50962,58 = 1,237P$$

$$\frac{50962,58}{1,237} = P$$

$$P \approx R41197,43$$

OR/OF

✓24

A

✓ 1,145P

CA

✓6

A

✓1,237P

CA

✓P the subject/die onderwerp

CA

✓ value/waarde

CA**OR/OF**

$$P_1 = \frac{50962,58}{\left(1 + \frac{5,20\%}{4} \right)^6}$$

$$= R 47162,27$$

$$\therefore P_2 = \frac{47162,27}{\left(1 + \frac{6,78\%}{12} \right)^{24}}$$

$$\approx R 41197,43$$

OR/OF

✓6

A✓ $\left(1 + \frac{5,20\%}{4} \right)$ **A**

✓ R 47162,27

CA

✓24

A✓ $\left(1 + \frac{6,78\%}{12} \right)$ **A**

✓ value/waarde

CA**NPR**
(6)**[14]**

QUESTION/VRAAG 6

6.1	$f(x) = 5 + x$ $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ $= \lim_{h \rightarrow 0} \frac{5 + x + h - (5 + x)}{h}$ $= \lim_{h \rightarrow 0} \frac{5 + x + h - 5 - x}{h}$ $= \lim_{h \rightarrow 0} \frac{h}{h}$ $= \lim_{h \rightarrow 0} (1)$ $\therefore f'(x) = 1$	✓ definition/definisie A ✓ Substitution/Verv. CA ✓ S CA ✓ S CA ✓ 1 CA Penalty of one mark for incorrect notation <i>Penaliseer een punt indien notasie foutief is.</i> AO: zero marks/punte (5)
6.2.1	$y = x(x+9)$ $= x^2 + 9x$ $\frac{dy}{dx} = 2x + 9$	✓ expansion/ontwikkeling A ✓ $2x$ CA ✓ 9 CA (3)
6.2.2	$D_x[\sqrt[7]{x} + \pi p^3]$ $= D_x\left[x^{\frac{1}{7}} + \pi p^3\right]$ $= \frac{1}{7}x^{-\frac{6}{7}}$	✓ $x^{\frac{1}{7}}$ A ✓✓ $\frac{1}{7}x^{-\frac{6}{7}}$ CA (3)
6.2.3	$f(x) = \frac{1-x^9}{x^2}$ $= x^{-2} - x^7$ $f'(x) = -2x^{-3} - 7x^6$	✓ x^{-2} A ✓ $-x^7$ A ✓ $-2x^{-3}$ CA ✓ $-7x^6$ CA (4)
6.3.1	$g(x) = -4x^2$ $g(2) = -4(2)^2 = -16$	✓ -16 A (1)

6.3.2	$g'(x) = -8x$ $m = g'(2) = -8(2)$ $m = -16$ $y + 16 = -16(x - 2)$ OR/OF $-16 = -16(2) + c$ $\therefore c = 16$ $\therefore y = -16x + 16$	✓ derivative/afgeleide A ✓ gradient/gradiënt CA ✓ substitution/verv. CA ✓ equation/verg. CA (4)
		[20]

7.5	$g(x) = -x^3 + 2x^2 + 5x - 6$ $g'(x) = -3x^2 + 4x + 5$ $-3x^2 + 4x + 5 = 0$ $x = \frac{-4 \pm \sqrt{(4)^2 - 4(-3)(5)}}{2(-3)}$ $x = 2,12 \text{ or/of } x = -0,79$ $g(2,12) = -(2,12)^3 + 2(2,12)^2 + 5(2,12) - 6$ $= 4,06$ $g(-0,79) = -(-0,79)^3 + 2(-0,79)^2 + 5(-0,79) - 6$ $= -8,21$ $E(-0,79; -8,21) \text{ and/en } F(2,12; 4,06)$	✓ derivative/afgeleide CA ✓ equating derivative to 0/ stel afgeleide gelyk aan 0 A ✓ SF CA ✓ both values of/beide waardes van x CA ✓ both values of /beide waardes van y CA NPR (5)
7.6	$x < -2$ or/of $1 < x < 3$ OR/OF $x \in (-\infty; -2)$ or/of $x \in (1; 3)$ OR/OF $x < -2$ or/of $x > 1$ and /en $x < 3$	✓ $x < -2$ CA ✓ critical values/kritiese waardes CA ✓ correct notation/korrekte notasie A OR/OF ✓ $x \in (-\infty; -2)$ CA ✓ critical values/kritiese waardes CA ✓ notation/notasie A OR/OF ✓ $x < -2$ CA ✓ critical values/kritiese waardes CA ✓ correct notation/korrekte notasie A (3) [15]

QUESTION/VRAAG 8

8.1	$h(t) = -5t^2 + 25t$ $h(1) = -5(1)^2 + 25(1)$ $= 20 \text{ m}$	✓ substitution/verv. ✓ height/hoogte A CA NPU (2)
8.2	$h'(t) = -10t + 25$ $h'(0) = -10(0) + 25$ $= 25 \text{ m/s}$	✓ $-10t + 25$ A ✓ substituting/verv. 0 ✓ initial velocity/aanvangsnelheid CA CA NPU (3)
8.3	$-10t + 25 = 0$ $10t = 25$ OR/OF $t = -\frac{b}{2a} = -\frac{(25)}{2(-5)}$ $t = \frac{5}{2} = 2,5 \text{ s}$ $t = 2,5 \text{ s}$ $h(2,5) = -5(2,5)^2 + 25(2,5)$ $= 31,25 \text{ m}$ OR/OF $h = \frac{4ac - b^2}{4a}$ $h = \frac{4(-5)(0) - (25)^2}{4(-5)}$ $h = 31,25 \text{ m}$	✓ $h'(t) = 0$ or/of SF A ✓ value of/waarde van t CA ✓ max. height/maks. hoogte CA OR/OF ✓ F A ✓ SF A ✓ max. height/maks. hoogte CA NPU Note: The 3rd mark (CA) should not be awarded if the t-value is not calculated using optimisation. Nota: Die 3^{de} punt (CA) moet nie toegeken word indien die waarde van t nie vanuit optimalisering. (3)

8.4	$h(t) = -5t^2 + 25t$ $30 = -5t^2 + 25t$ $0 = -5t^2 + 25t - 30$ $0 = t^2 - 5t + 6$ $0 = (t - 2)(t - 3) \text{ OR / OF } t = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(1)(6)}}{2(1)}$ $t = 2\text{s or / of } t = 3\text{s}$	✓ equating to/gelykstelling aan 30 A ✓ factors/ faktore or/of SF CA ✓ both values of/beide waardes van t CA NPU (3) [11]
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QUESTION/VRAAG 9

9.1.1	$\int (10^x + 6) dx$ $= \frac{10^x}{\ln 10} + 6x + C$	✓ $\frac{10^x}{\ln 10}$ ✓ $6x$ ✓ C	A A A (3)
9.1.2	$\int (x^4(x + 2) - 2x^{-3}) dx$ $= \int (x^5 + 2x^4 - 2x^{-3}) dx$ $= \frac{x^6}{6} + \frac{2x^5}{5} + x^{-2} + C$ Note: No penalty if C is omitted <i>Nota: Geen penaliseering indien C weggelaat is</i>	✓ $x^5 + 2x^4$ ✓ $\frac{x^6}{6}$ ✓ $\frac{2x^5}{5}$ ✓ x^{-2}	A CA CA A (4)
9.2	$A = - \int_2^4 \left(\frac{3}{x} - 4 \right) dx$ $= - (3 \ln x - 4x) \Big _2^4$ $= - \left[- [3 \ln(4) - 4(4)] - [3 \ln(2) - 4(2)] \right]$ $\approx 5,92 \text{ square units/vierkante eenhede}$ OR/OF $A = \int_2^4 \left(\frac{3}{x} - 4 \right) dx$ $= (3 \ln x - 4x) \Big _2^4$ $= [3 \ln(4) - 4(4)] - [3 \ln(2) - 4(2)]$ $\approx -5,92$ $\therefore A \approx 5,92 \text{ square units/ vierkante eenhede}$	✓ Area notation using integrals/ Oppervlakte-notasie met gebruik van integrale ✓ $3 \ln x$ ✓ $-4x$ ✓✓ SF ✓ Shaded area/gearseerde oppervlakte CA OR/OF ✓ Area notation using integrals/ Oppervlakte-notasie met gebruik van integrale ✓ $3 \ln x$ ✓ $-4x$ ✓✓ SF ✓ positive shaded area/positiewe gearseerde oppervlakte CA CA NPU NPR (6) [13]	

TOTAL/TOTAAL: 150